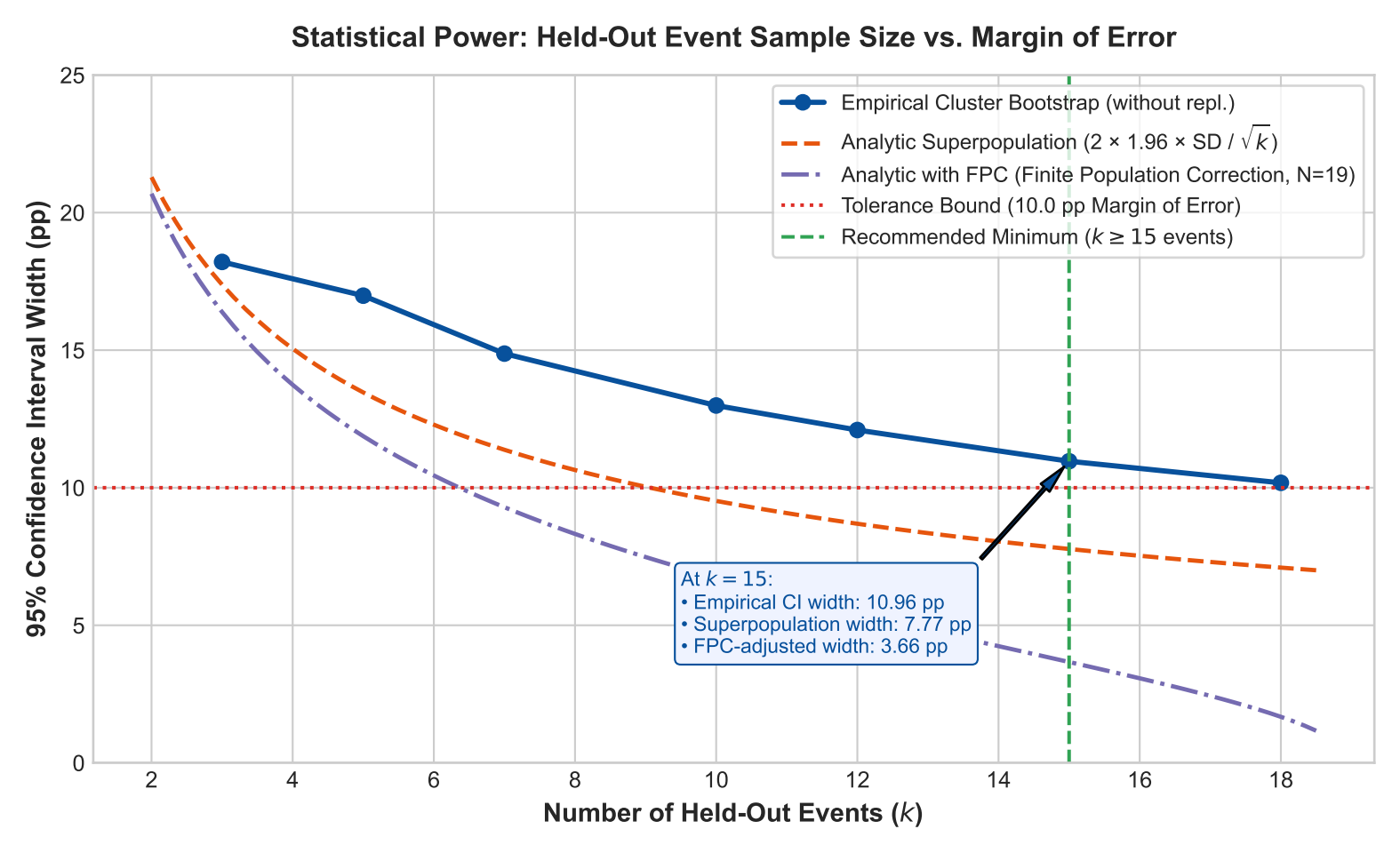
Statistical Power: Held-Out Event Sample Size vs. Margin of Error
95% Confidence Interval Width (pp)
Number of Held-Out Events (k)
0
5
10
15
20
25
2
4
6
8
10
12
14
16
18
Empirical Cluster Bootstrap (without repl.)
Analytic Superpopulation (2 × 1.96 × SD / √k)
Analytic with FPC (Finite Population Correction, N=19)
Tolerance Bound (10.0 pp Margin of Error)
Recommended Minimum (k ≥ 15 events)
At k = 15:
• Empirical CI width: 10.96 pp
• Superpopulation width: 7.77 pp
• FPC-adjusted width: 3.66 pp